Fiscal Year:	FY 2020	**Task Last Updated:**	FY 10/23/2019
PI Name:	Ajaev, Vladimir Ph.D.		
Project Title:	Modeling Phase Change and Thermocapillary Flow in the Constrained Vapor Bubble Experiment		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Fluid physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	75275-0156	**Congressional District:**	32
Comments:			
Project Type:	Ground,Physical Sciences Informatics (PSI)	**Solicitation / Funding Source:**	2015-16 Physical Sciences NNH15ZTT001N-15PSI-C: Use of the NASA Physical Sciences Informatics System – Appendix C
Start Date:	12/22/2017	**End Date:**	12/21/2020
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
Contact Monitor:	Chao, David	**Contact Phone:**	216-433-8320
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 12/21/2020 per NSSC information (Ed., 11/26/19)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Plawsky, Joel Sc.D. (Rensselaer Polytechnic Institute)		
Grant/Contract No.:	80NSSC18K0332		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	We propose theoretical and numerical studies of constrained vapor-liquid interfaces in the presence of phase change, i.e., evaporation and/or condensation. These studies will result in better understanding of heat and mass transfer in two-phase systems in which the effects of capillary forces and thermocapillarity are important. The main focus of the work is on a specific geometric configuration used in a recent space experiment, referred to as a Constrained Vapor Bubble (CVB). We propose to use asymptotic methods to simplify the system of governing equations for fluid flow and heat transfer for this configuration, followed by numerical solutions of the resulting equations using modern computational techniques. The proposed work has a wide range of ground and space applications including two-phase cooling systems for traditional electronic devices such as laptop computers as well as novel thermal management systems for emerging applications such as proton exchange membrane fuel cells and photovoltaic cells for solar energy harvesting. The small length scales in many of these systems imply that interfacial phenomena are very important while the effects of gravity are secondary. Some aspects of the proposed work are also relevant for modeling of physical processes such as self-assembly of colloidal particles from evaporating droplets, condensation of droplets in novel devices for energy conversion, and water desalination. The liquid transport mechanisms by capillary forces are also important to study for space applications, such as advanced cooling systems for spacecraft, since conventional methods for liquid transport can fail or become impractical in such applications. For example, any system that relies on the force of gravity for functioning is likely to fail in the microgravity environment. Transporting liquids by external pressure gradients, while not impossible in space, typically requires mechanical pumps, often making such systems impractical because of the payload weight restrictions.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The proposed work has a wide range of ground and space applications including two-phase cooling systems for traditional electronic devices such as laptop computers as well as novel thermal management systems for emerging applications such as proton exchange membrane fuel cells and photovoltaic cells for solar energy harvesting.
Task Progress:	Significant progress has been made in modeling and simulation of droplet formation at the walls of the cuvette in the constrained vapor bubble system by the Principal Investigator (PI) and his graduate student at Southern Methodist University (SMU). A model of axisymmetric droplet is developed which accounts for heat conduction in the walls. The simulation results are shown to agree with the experimental data from J. Plawsky's group. A non-axisymmetric formulation is developed to describe the central droplet formed in the microgravity experiments. Simulations of corner flows are conducted, which account for phase change at the liquid surface and temperature variation along the cuvette. The focus of the work at Rensselaer Polytechnic Institute (RPI) was in analyzing some of the data from CVB-1 and CVB-2 and doing some ground-based experiments that will help us eventually understand the difference between the two experiments.
Bibliography Type:	Description: (Last Updated: 06/13/2025)
Abstracts for Journals and Proceedings	Ajaev VS, Barrett JA. "Shapes and dynamics of confined liquid-vapor interfaces in two-phase systems in microgravity." 26th ELGRA Biennial Symposium and ITTW Conference, Granada, Spain, September 24-27, 2019. Book of Abstracts, 26th ELGRA Biennial Symposium and ITTW Conference, Granada, Spain, September 24-27, 2019. , Sep-2019
Abstracts for Journals and Proceedings	Barrett JA, Ajaev VS. "Modeling of moving liquid-vapor interfaces in the constrained vapor bubble system." Bluebonnet Texas Fluid Mechanics Symposium, Dallas, TX, April 2019. Book of Abstracts. Bluebonnet Texas Fluid Mechanics Symposium, Dallas, TX, April 2019. , Apr-2019
Abstracts for Journals and Proceedings	Nguyen T, Plawsky JL, Wayner PC. "Disjoining pressure: redefining evaporation." SHTC 2019. ASME Summer Heat Transfer Conference, Bellevue, WA, July 14 – 17, 2019. Program and Abstracts. SHTC 2019. ASME Summer Heat Transfer Conference, Bellevue, WA, July 14 – 17, 2019. , Jul-2019
Articles in Peer-reviewed Journals	Barrett JA, Ajaev VS. "Modeling of moving liquid-vapor interfaces in the constrained vapor bubble system." Microgravity Science and Technology. 2019 Oct;31(5):685-92. https://doi.org/10.1007/s12217-019-09741-7 , Oct-2019